MTH 732 Topology - Homework 4

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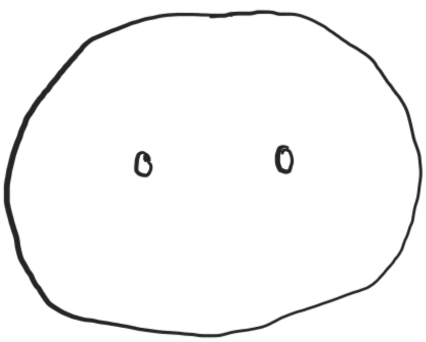
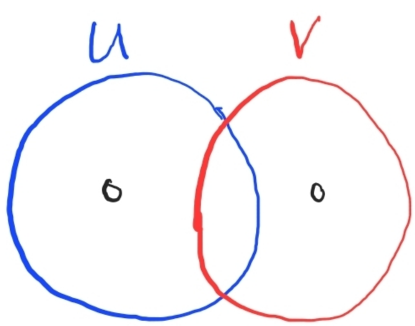
Assignment 1. Compute the fundamental group of $\mathbb{R}^2$ with two points removed.

Proof. (Note: all disk boundaries pictured in this proof are understood to be open.)
We can view this homeomorphically as $\text{int}(D^2)$ with two points removed:



We can break this up into two open disks U and V, each with one point removed:



We know that $\pi_1(U) = \langle a \rangle$ and $\pi_1(V) = \langle b \rangle$ while $\pi_1(U \cap V)$ is trivial. Then we can use Seifert-van Kampen to get

$$\pi_1(U \cup V) = \langle a, b \rangle \, .$$

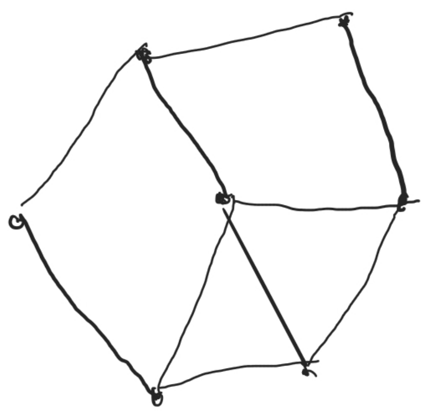
$\square$

Assignment 2. If $F \subset \mathbb{R}^3$ is a finite set, show that $\mathbb{R}^3 \setminus F$ is simply-connected.

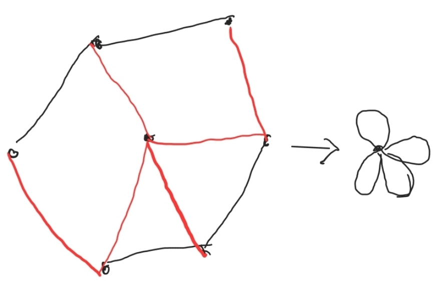
Proof. Let n be the size of F, and define $F_n = \{(0,0,0), \dots, (n-1,0,0)\}$. Also, define $B^3 = \text{int}(D^3)$ centered at the origin. We can then model $\mathbb{R}^3 \setminus F$ homeomorphically as $\mathbb{R}^3 \setminus F_n$. Clearly $\mathbb{R}^3 \setminus F_n$ is path-connected, so we will focus on the fundamental groups.

First, when $n = 1$, $\mathbb{R} \setminus F_n$ has S^2 as a deformation retract, which we know to have trivial fundamental group. The same is true for $B^3 \setminus F_n$ in this case since it is also a deformation retract of $\mathbb{R}^3$. Now assume we have proven the hypothesis through $n-1$, and define $C = \{(x,y,z) : x^2+y^2+z^2 > 0.9\}$. Then we can express $\mathbb{R}^3 \setminus F_n$ as a union of $B^3 \setminus F_1$ with $C \setminus F_{n-1}$. Luckily, $C \setminus F_{n-1}$ is homotopy equivalent to $B^3 \setminus F_{n-1}$, and since $(B^3 \setminus F_1) \cap (C \setminus F_{n-1})$ is homotopy equivalent to S^2, we know everything we need for Seifert-van Kampen. Since all the fundamental groups in question are trivial, so is $\pi_1(\mathbb{R}^3 \setminus F)$. □

Assignment 3. Compute the fundamental group of the following graph:



Proof. Consider the deformation retract pictured, where the red spanning tree is contracted to a point:



Thus the graph has the same fundamental group as a wedge of four circles, which is the free group on four generators. □

Assignment 4. Let X be the union of n lines through the origin in $\mathbb{R}^3$. Compute the fundamental group of its complement $\pi_1(\mathbb{R}^3 - X)$.

Proof. Note that since, in particular, the origin is not in $\mathbb{R}^3 - X$, we can deformation retract this space to the unit sphere with $2n$ points removed. We can compute the fundamental group, then, by expressing it as the union of two punctured open discs U and V, say where $z > -0.1$ and $z < 0.1$ respectively. Furthermore, we can consider $U \setminus V$ to contain exactly one of the points and $V \setminus U$ to contain all $2n-1$ others (which can be achieved by homeomorphism).

First, note that $U \cap V$ is the "open band" or annulus $S^2 \cap \{(x, y, z) : -0.1 < z < 0.1\}$, so $\pi_1(U \cap V)$ is the free group on one generator γ. Also, $\pi_1(U)$ is the free group on one generator, say α, and $\pi_1(V) = \langle \beta_1, \ldots, \beta_{2n-1} \rangle$. Finally, the image of γ induced by inclusion into U and V is α and $\beta_1 * \cdots * \beta_{2n-1}$ respectively, since it represents a loop running along/near the outer edges of U and V. Seifert-van Kampen tells us, then (since all of these are path-connected), that

$$\begin{aligned} \pi_1(\mathbb{R}^3 - X) &= \pi_1(U \cup V) \\ &= \langle \alpha, \beta_1, \ldots, \beta_{2n-1} : \alpha = \beta_1 * \cdots * \beta_{2n-1} \rangle \\ &= \langle \beta_1, \ldots, \beta_{2n-1} \rangle . \end{aligned}$$

Thus $\pi_1(\mathbb{R}^3 - X)$ is the free group on $2n - 1$ generators. □

Assignment 5. Compute the fundamental group of the connected sum

$$\mathbb{RP}^2 \# \cdots \# \mathbb{RP}^2$$

of k copies of the projective plane $\mathbb{RP}^2$.

Proof. As we have done before, we can use Seifert-van Kampen recursively. First, when $k = 2$, we can take neighborhoods U and V of the two copies of $\mathbb{RP}^2$ which intersect only on a small neighborhood of the disk along which they are connected. That intersection will have free fundamental group on one generator. Taking the model of $\mathbb{RP}^2$ as identifying antipodal points in S^2, we need to consider taking an open disk out, which is what happens when we take the connect sum. If we take a disk out, we can think of each copy of $\mathbb{RP}^2$ as, say, $U = S^2 \cap \{(x, y, z) : z > -0.1\}$ and $V = S^2 \cap \{(x, y, z) : z < 0.1\}$ under the appropriate identifications. Thus we can consider the generator of $\pi_1(U \cap V)$ as the equator ($z = 0$), which we know can be expressed as a^2 in $\pi_1(U)$ and b^2 in $\pi_1(V)$, where a and b are the usual projections of the "halfway around the equator" path. Thus

$$\pi_1(\mathbb{RP}^2 \# \mathbb{RP}^2) = \left\langle a, b : a^2 = b^2 = 1 \right\rangle .$$

Repeating this calculation for a general k through induction, we get

$$\pi_1(\mathbb{RP}^2 \# \cdots \# \mathbb{RP}^2) = \left\langle x_1, \ldots, x_k : x_1^2 = \cdots = x_k^2 = 1 \right\rangle .$$

□